

TOTAL SCOPE: 60 HOURS • \$3,000.00

Milestone	Technical Deliverables & Architecture	Timeline	Total
Milestone 0 Prototype	Interactive Working Proof-of-Concept (Delivered Upfront) Working Next.js 15 reference cockpit deployed at docref-vault.vercel.app with split-screen OCR bounding boxes, dual-provider AI query execution, and dynamic 4-role RBAC switcher.	Shipped	\$0.00
Milestone 1	Postgres pgvector Foundation, Ingestion Engine & OCR Parsing Production Postgres schema with pgvector (1536-dim HNSW index) and GIN index. Asynchronous document ingestion with OCR bounding-box coordinate extraction ([ymin, xmin, ymax, xmax]) and chunk SHA-256 sealing.	18 hrs	\$900.00
Milestone 2	Hybrid RAG Retrieval, Cross-Encoder Re-Ranking & AI Gateway Dense vector similarity combined with sparse lexical BM25. Two-stage cross-encoder re-ranking pipeline. Dual-provider AI gateway (OpenAI gpt-4o-mini + Gemini 2.0 Flash + deterministic local fallback).	16 hrs	\$800.00
Milestone 3	Split-Screen Cockpit UI, RBAC Security Gate & Deterministic Reports Interactive split-screen reference workspace with bi-directional citation highlighting. 4-role RBAC authorization gateway. Cryptographic deterministic report generator with SHA-256 seal.	14 hrs	\$700.00
Milestone 4	Anti-Abuse Shield, WORM Audit Logs, Security Hardening & Handoff Token bucket rate limiting (60 req/min), canary token injection trap, immutable WORM audit logs. Comprehensive handoff package: OpenAPI 3.1 specs, Docker blueprints, and administrator runbooks.	12 hrs	\$600.00
TOTAL FIXED INVESTMENT (TURNKEY ARCHITECTURE & PRODUCTION HANDOFF)		60 hrs	\$3,000.00

Hybrid RAG Pipeline	pgvector HNSW (Dense) + BM25 (Sparse)
Re-Ranking Engine	Cross-Encoder Rescoring (Dense 0.55 / BM25 0.45)
OCR Coordinate Citations	Exact [ymin, xmin, ymax, xmax] Bounding Box
Cryptographic Verification	SHA-256 Deterministic Attestation Seal

- ✓ Zero Black-Box AI: Every synthesis cites verified source document chunks.
- ✓ Anti-Abuse Shield: Token bucket rate limiter (60 req/min) + canary injection quarantine.
- ✓ Dual AI Provider: OpenAI gpt-4o-mini + Gemini 2.0 Flash sub-second failover.
- ✓ RBAC Hierarchy: 4-role permission gating across UI and API endpoints.

3-4 weeks total delivery with functional bi-weekly staging deploys for continuous testing.

100% full intellectual property transfer upon milestone sign-off. Clean MIT-licensed repo.

30 days of post-handoff bug fixes and developer support included at zero additional cost.

Docker compose configs, pgvector migrations, OpenAPI specs. and deployment runbooks.

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Signatory: Shakil Ahmed, Founder & Principal Systems Architect

Signatory: Authorized Representative / Engineering Lead

AUTHORIZED CONTRACTOR SIGNATURE _____

AUTHORIZED CLIENT SIGNATURE _____